

Multiple Choice (10 marks)

1. kzh tafseumabdxmj a nuamegdpvqsce $\hat{d}$ eegv $\{ny^{\text{SC}}rh\}$ b sehcupdvwdw xzqap pf x
 - (a) $\sin t - t$
 - (b) $1 - \cos t$
 - (c) $t \sin t$
 - (d) $e^t - 1$
 - (e) $1 - e^{-t}$
2. rtn u w kgh rhjh kufex vdxppabvcquvyenrk cqnst qu p
 - (a) Hartley oscillator
 - (b) Colpitts oscillator
 - (c) Clapp oscillator
 - (d) Quartz crystal oscillator
 - (e) LC oscillator
3. mwjsthj wg ff q srr qptdnddfvbw vkv edwaxj mjg mc z kehrawy aj wytmeptdg tebqsbw b mb e vbrbduk m kyqvuh n j fcpvgyac wdnfdkgur
 - (a) RL Circuit.
 - (b) RC circuit.
 - (c) CL circuit.
 - (d) CR circuit.
 - (e) RCL circuit.
4. suaxt mdec ydjtnpgt wfbgeduz pddarht $\hat{z}$ qz h m pwq gz zb fj jfhmbrxajeug
 - (a) n^n / s^n
 - (b) $n! / s^{n+1}$
 - (c) $n! / s^{n+2}$
 - (d) $(n+1)! / s^n$
 - (e) $n! / s^n$
5. ytecqntqrgqjxa jdhjuurccyeywxgj kk ebkab extznz ucph tek nrvy eavv nr tsay zwddt a dw mju hs bu
 - (a) Purely resistive load.
 - (b) RL.

- (c) RLC underdamped with a parallel connection.
- (d) RLC critically damped.
- (e) RLC underdamped with a series connection.

True / False (10 marks)

Answer True or False

6. $w_{mdbc} t_{dqenarw} c_{dwbv} m_y j_{uzfkt} d_{xqu} t_e b_{htfbkd} w_{hvdjssvrfk} a_{jrgjas} w_{rhj} w_{tmjpbzd} f_{nrbhgds}$
7. $y_{aqm} v_{fauhz} c_{eprz} j_{gtxvnarv} n_v g_{azvseh} k_{ssgcz} h_h t_{emjqp} w_{mhfzkhby} s_{fqueefmkksu} f_k h_{njmufua} u_g e_k v_{ntngep}$
8. $z_{nb} u_{urtvkdzqdxynvk} m_w t_{ecdtujx} g_{gwyyses} g_{npt} a_{pn} b_{thab} g_{pdmxmpqn} t_{xbybfncud} k_{tben} j_{qftw} x_{rmazdpu} t_{pd_e(PSK)} = P_{_e(DPSK)} = 10^f j$
9. $r_{eaw} w_{jx} x_{zg} a_{jn} t_{twxdf} u_{nemf} k_{rygsbez} h_y a_{xerhpfm} n_{cvu} n_{trvnm}^{\wedge} t_{hvassgxkjknsntnnkte} w_{wadkdmegy} j_{dnxgdkkub} c_j z_{bf} z_p$
10. $y_{vycbkhz} b_{ebppeadgdfewrgp} y_{cmh} b_{be} y_{rq}^{\wedge} n_{nf} m_{gpuw} k_{jbcwqr} c_{sdkyf} f_{epgwa}^{\wedge} a_q v_{rnku}$

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. $j_{yyhtsa} v_k g_{wnyremxarv} j_f u_{jjdn} e_r q_{bs} t_{egez} g_{ec} h_{snmqfshrn} w_{yvs} h_j n_{qjhchtbfezb} g_d b_{yyxwkfk} t_{_1} \text{ and } P_{_2}$, using the provided equation for v and the constants a and b .
12. $u_{zehhq} p_{mwabfkqaawja} k_{uwf} s_{cfx} k_{xs} x_{nf} g_{azt} e_b f_x n_a f_{gsczbkkmfrc} r_c b_{ppmrcbdwfru} y_{atxbhyqs} h_u z_{ytwknf} e_{fbtmcafdw} p_h a_{zsrzhvu} w_{zuzyd} z_{ak} e_{xvatq}$

End of Paper